# Biochemistry Cume

**March 26, 2004**

1. (30pts) Consider the following peptide: WAYNSWRLD. What is the approximate net charge on the peptide at pH 1, 8 and 10?

2. (30pts) Determine the sequence of a peptide consisting of 14 amino acids on the basis of the following data.

   Amino acid composition: (A,D,2G,2K,3L,2N,Q,S,T)
   N-terminal analysis: N
   Carboxypeptidase digestion: Q
   Trypsin digestion: (A,G,Q), (D,G,2N,L,K), (S,T,2L,K)
   N-terminal analysis of (AGQ): A
   Chymotrypsin digestion: (K,L), (T,L), (A,G,S,K,Q) (D,L,G,2N)
   Staph. Protease (A,2K,3L,S,T,G,Q), (2N,D,G)
   Hydroxylamine: N, (A,D,2G,2K,3L,N,Q,S,T)

3. (20pts) Draw the hydrogen-bonding pattern that water forms with the Asn residue side chain at pH 2.

4. (20pts) Which amino acid **residue(s)** have two chiral centers? [Either give the name(s) or draw the structure(s).]

Possibly usseful information

| Side Group | | pKa |
|---|---|---|
| Arginine | R | 12.5 |
| Aspartic acid | D | 3.9 |
| Cysteine | C | 8.3 |
| Glutamic acid | E | 4.1 |
| Histidine | H | 6.0 |
| Lysine | K | 10.1 |
| Tyrosine | Y | 10.1 |
| C-terminus | | 2.5 |
| N-terminus | | 9.0 |

Specific cleavage of polypeptides.

| Reagent | Cleavage site |
|---|---|
| Carboxypeptidase | amino side of C-terminal amino acid |
| Trypsin | carboxyl side of lysine and arginine |
| Chymotrypsin | carboxyl side of tyrosine, tryptophan, phenylalanine, leucine, and methionine |
| Staph protease | carboxyl side of aspartate and glutamate residues. |
| Hydroxylamine | asparagine-glycine bonds |